

Notes 2

1. What is an Operating System?

An operating system provides all fundamental software features of a computer. An OS enables you to use the computer's hardware providing you the basic tools that make the computer useful. All of those features rely on the OS's kernel. Other OS features are owed to additional programs that run atop the kernel.

2. What is a kernel?

An OS kernel is a software component that's responsible for managing low-level features of the computer, including the following managing system hardware, memory allocation, CPU time, and program to program interaction.

3. Which other parts aside from the kernel identify an OS?

The other parts aside from the kernel identify an OS are:

- **Command-Line Shells:**
 - This was the de facto way of using computers before the Graphical Interface was invented. CMDs work by typing commands in a shell. In Linux, the entire system can be control via the CLI.
- **Graphical User Interfaces (GUI):**
 - GUIs rely on icons, menus, and a mouse pointer for the user interaction. Linux relies on a GUI known as the X Window System in combination with desktop environment program suites.
- **Utility and Productivity Programs:**
 - Tools like web browsers, documents processors and text editors.
- **Libraries:**
 - Libraries are collections of programming functions that can be used by a variety of programs.

4. What is Linux?

- Linux is a **Unix-like Operating System** popular in academic and business environments.
- Linux consists of a **kernel, libraries, and utilities** that make up the entire operating system.
- Linux is available in many **distributions***(any operating system that runs the Linux kernel)
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5. What is a Linux distribution?

- Linux distributions are:
 - **Arch**
 - **CentOS**
 - **Debian**
 - **Fedora**
 - **openSUSE**
 - **Red Hat**
 - **Slackware**

- **Ubuntu**

6. List at least 4 Linux characteristics:

- 4 Linux characteristics are:
 - Linux **open source software**
 - Linux is available **free of charge**
 - Linux includes many of the **Unix tools** including many important Internet server programs and programming languages out of the box
 - Linux is **highly scalable** and customizable

7. What is Debian?

Debian is an **all-volunteer organization** dedicated to developing free software and promoting the ideals of the Free Software community.

8. List and define the different types of licensing agreements

- Types of licensing agreement:
 - **Open Source:**
 - the software may be distributed for a fee or free. The source code is distributed with the software.
 - **Cloud Source:**
 - the software is not distributed with the source code. The user is restricted from modifying the code.
 - **Free:** the software is free but the source code is not available.
 - **Shareware:** the software is free on a trial basis.
 - **Free software:**
 - the software is distributed with the source code. The software can be free of charge or obtained by a fee.

9. What is Free Software? Define the 4 freedoms.

- **Free software:**
 - the software is distributed with the source code. The software can be free of charge or obtained by a fee.
- **The 4 freedoms:**
 - **Freedom 0:**
 - use the software for any purpose
 - **Freedom 1:**
 - examine the source code and modify it as you see fit.
 - **Freedom 2:**
 - redistribute the software
 - **Freedom 3:**
 - redistribute your modified software

10. What is virtualization?

- **Virtualization** is defined as creating virtual versions of something.
- Virtualization is often used to let **multiple OSs run on one physical machine** at the same time.
- Virtualization allows administrators to **divide the hardware** and create multiple computers **inside a single physical computer**.